Literature review

Does the pretraining corpus matter for molecular property prediction? MIT 6.3930, autumn 2026. Gabriel Dupuis, Léonard Mauvernay. Compiled 8 November 2026 from the reading list in `papers/`.

Every paper below was read in full. Each entry says what the authors did, the numbers that matter to us, and what it changes in our design. Section 10 is the list of changes; if you read nothing else, read that.

1. Why this review exists

Our project holds CheMeleon's architecture, objective and training budget fixed and varies only the pretraining corpus, first in size, then in composition. Two questions decide whether that is worth doing:

1. Does pretraining help at all on small molecular property tasks?
2. If it helps, is the corpus the lever, or is it the objective?

The literature answers the first with "barely, and inconsistently" and has never properly tested the second. That gap is our project.

2. The foundation we build on

Chemprop (Heid et al., JCIM 2024)

The D-MPNN implementation and our from-scratch baseline. A software paper: all benchmark numbers live in the supplement, which we could not retrieve, so cite it for architecture and defaults only, never for performance numbers.

Defaults that we must lock across every arm: hidden size 300, three message-passing steps, mean aggregation, two-layer FFN of 300, Adam with linear warm-up 1e-4 to 1e-3 over two epochs then exponential decay to 1e-4, batch 50, thirty epochs, random split.

Two facts shape our baselines. First, extra descriptors enter after aggregation and before the FFN, so "Chemprop + descriptors" is a one-flag change rather than a different codebase. Second, modules one to three can be bypassed entirely so that the molecular embedding *is* the descriptor vector, which gives us a fourth baseline for free: a plain feed-forward network on Mordred descriptors with no graph at all. That baseline isolates how much of CheMeleon's advantage is descriptor information rather than descriptor-shaped pretraining, which is very close to our actual question.

The paper also warns that "small data sets need a much larger number of epochs to train". Thirty epochs will under-train anything at BBBP or ESOL scale.

> ***Caution.*** *The file we downloaded as the Chemprop paper was the wrong article entirely; `arxiv.org/abs/2404.01926` is an astrophysics preprint about pulsars, and Chemprop was never posted to arXiv. The reading list's link needs correcting. We replaced it with the open-access text from PubMed Central.*

CheMeleon (Burns, Zalte et al., JCIM 2026)

Our fixed recipe. A D-MPNN pretrained to regress 1,613 Mordred descriptors over one million PubChem molecules drawn uniformly at random, with 85 percent of the descriptor targets masked in the loss on each example, targets z-scored and Winsorised at six standard deviations. Message-passing width 2048, depth 6, mean aggregation, 8.7 million encoder parameters. Fine-tuning discards the pretrained head and trains everything end to end.

Results: wins 75 percent of 28 Polaris and TDC tasks against minimol at 71, random forest at 68 and

from-scratch Chemprop at 32; wins 97 percent of 30 MoleculeACE assays where from-scratch Chemprop wins none. Pretraining lifts the *same architecture* from 32 to 75 percent, and a size-matched randomly initialised model is much worse, so the gain is pretraining rather than capacity.

What matters most for us is what the paper does **not** do. One corpus, one size, one composition, one descriptor set, no scaling analysis, no ablation of the corpus, and no check for overlap between the PubChem corpus and the benchmarks. The authors explicitly park the question of whether the pretraining loss value even predicts downstream performance: it is "a question left for future work". Stages C and D are therefore a genuine gap, not a replication.

Three design consequences. The corpus is an unfiltered random PubChem draw with no deduplication or standardisation, so any cleaning we add is a second variable, not a neutral improvement. The descriptor standardisation is computed over the whole corpus, so at 500 or 1,000 molecules those statistics are unstable: freeze the column set and the scaler from the released one-million store and reuse them unchanged at every size. And with 85 percent of targets masked, a small corpus sees each descriptor very few times, so we should report the effective supervised signal (molecules times 0.15 times 1,613) next to the molecule count.

One caveat worth repeating in the write-up: the paper reports that CheMeleon maps structurally similar molecules into the same region of feature space, which *hurts* activity-cliff discrimination. Stage E should therefore expect the relationship between corpus similarity and gain to be non-monotonic.

3. Does pretraining help at all?

Hu et al., ICLR 2020 - the founding negative result

The reference on negative transfer. GIN, scaffold split, ten seeds, pretraining on 2 million ZINC15 molecules (self-supervised) plus 456,000 ChEMBL molecules with 1,310 assays (supervised).

Node-level or graph-level pretraining alone gains only three to four ROC-AUC points on average and causes **negative transfer on two of eight datasets**. Only the combination of node-level and graph-level pretraining avoids negative transfer everywhere, for +7.2 points. Gains concentrate on the smallest datasets: ClinTox +14.6, BACE +14.4, both around 1,500 molecules.

Two numbers to carry into our design. Non-pretrained seed standard deviations are ±4.5 on BBBP, ±5.4 on BACE and ±4.4 on ClinTox, so a corpus difference of one to three points is invisible without paired seeds and many of them. Pretraining *reduces* that variance (BACE from ±5.4 to ±0.7), which makes variance reduction a corpus-quality signal in its own right. They also removed all downstream test molecules from the supervised pretraining set, which is exactly the de-duplication our protocol is still arguing about.

> *"pre-training GNNs does not always help. Naive pre-training strategies can lead to negative transfer on many downstream tasks."*

Sun, Dai and Yu, NeurIPS 2022 - the methodological model

The paper our question is shaped like. One-factor-at-a-time ablation from a fixed baseline configuration, varying objective, input features, split, corpus scale, backbone and depth.

Self-supervised pretraining alone gains +0.12 to +0.13 average ROC-AUC against seed standard deviations of 1.2 to 1.6, which is to say nothing. Supervised ChEMBL pretraining gains +0.7 to +1.6. Rich input features alone add +1.3 to +3.1, more than any self-supervised objective. Most damning for method: with the default learning rate and a single split, pretraining looks clearly beneficial (BACE 79.5 to 84.0); sweeping seven learning rates and averaging three splits, the gain disappears entirely.

On corpus scale they compare ZINC15 at 2 million against SAVI at about 1 billion molecules, 500 times more

data, and conclude it "does not lead to a significant improvement". Pretraining accuracy exceeds 90 percent after 0.1 to 0.2 million steps, before a single epoch of the large corpus. Corpus *composition* is never varied. A flat size curve from 500 to one million molecules is therefore an entirely plausible outcome for us, and we should log the pretraining loss curve per size so that saturation is visible as a mechanism rather than asserted.

> *"hyper-parameters could have larger impacts on accuracy of downstream tasks than the choice of pretraining tasks, especially when the scales of downstream tasks are small."*

Deng et al., Nat. Commun. 2023 - the evaluation audit

62,820 models, 30 split seeds, scaffold and random splits, Mann-Whitney U tests throughout. Random forest on RDKit 2D descriptors beats frozen MolBERT and GROVER on BACE, BBBP, ESOL and Lipophilicity under scaffold split. Frozen pretrained models are competitive only below about one thousand training molecules, and GROVER never improves with more data at all.

Three findings change our plan directly.

ESOL is descriptor-saturated, not merely noisy. Its label correlates at roughly 1.0 with computed LogP, and a model using eleven physicochemical descriptors is top-tier on it. A corpus effect cannot show through that. This is a stronger reason to demote ESOL than the scaffold-block noise we had already measured.

Three splits can crown the wrong model. Across 30 scaffold splits, random forest is best on only 23 of 30 folds for BACE and 20 of 30 for BBBP. Many three-split subsets would give the opposite verdict. Our two seeds are not enough.

Activity cliffs dominate. Cliff molecules are 25 to 46 percent of molecules in activity datasets, and pretrained models essentially predict the mean for them. Aggregate metrics hide this, so test metrics should be split into cliff and non-cliff subsets.

They also show that conclusions flip with the metric: random forest beats all pretrained models on BBBP by AUROC but ties on precision and negative predictive value.

> *"a model cannot save an unqualified dataset which cannot remedy an improper evaluation for an ambiguous chemical space generalization claim."*

Praski, Adamczyk and Czech, arXiv 2508.06199 (2025)

28 pretrained embedding models frozen, 25 classification datasets, identical downstream heads, Bayesian Bradley-Terry analysis with a region of practical equivalence. **Only CheMeleon is rated better than the plain ECFP count fingerprint.** Nine models are practically equivalent to it and eighteen are worse.

The pattern they identify is the single most important result in this review for framing our project: the four neural models that rank above ECFP (CheMeleon, R-MAT, MolBERT, ChemBERTa) all share one thing, and it is not their corpus. It is their objective, multitask regression of computed molecular descriptors. Masking, contrastive, causal-language-model and 3D objectives all rank below the fingerprint.

On external validation, CheMeleon's advantage over ECFP is +0.65 to +0.74 percent AUROC, below their one-percent equivalence threshold, so the verdict is "inconclusive". If the best pretrained model beats a fingerprint by less than one percent, a corpus effect inside that model will be smaller still. Plan the statistics accordingly.

The paper gives no table of pretraining corpora at all and states plainly that it omits "pretraining dataset curation" details. Its conclusion that "relying solely on large pretraining datasets... is insufficient" is an inference from objective patterns, not a tested claim. Another gap we can fill.

The 1 October lecture (Barzilay, with Yuxin Xie's benchmarking)

The study that started this project, in 57 slides. Five pretrained models versus a well-tuned Chemprop on

roughly 45 datasets: 11 MoleculeNet, 22 TDC ADMET and 12 proprietary pharmaceutical sets, under "same splits, same tuning budget, same ensembles for every model".

Benchmark	Critical difference	Ranking (lower is better)
TDC ADMET, 22 endpoints	1.61	CheMeleon 2.14, MotiL 3.25, MoLFormer 3.27, Chemprop 3.34, MolFCL 3.64, MolCLR 5.36
MoleculeNet, 10 datasets	2.38	MotiL 2.80, MoLFormer 2.90, CheMeleon 3.10, MolFCL 3.70, Chemprop 3.80, MolCLR 4.70

The verdict slide reads: "No. Models joined by a thick bar are not significantly different, and Chemprop2 is always inside the bar with the top foundation models." On MoleculeNet the Friedman test gives p = 0.194: nothing differs from anything. Per-dataset Tukey tests on the pharmaceutical assays give "Chemprop2 is best or tied on most datasets", and on the one physicochemical regression task the from-scratch baseline wins outright.

The low-data sweep retrains every model on 5 to 70 percent of the training set. On BACE the foundation-model advantage is largest at 60 training molecules (about +0.07 AUROC) and essentially gone by 847. On ESOL, two models are catastrophically worse than from-scratch at 45 molecules (RMSE above 2.0 against 1.26) and never fully recover.

Four observations from those slides bear directly on us.

Corpus size is anti-correlated with rank across these six models. CheMeleon on 1 million PubChem molecules ranks first on ADMET; MolCLR on 10 million ranks last; MoLFormer on 1.1 billion sits mid-pack and is one of the two models that actively hurt on ESOL. Confounded with objective and backbone, of course, which is precisely why a controlled corpus experiment is worth running.

Descriptors matter as much as pretraining. Chemprop with descriptors beats Chemprop without by about 0.09 AUROC at 60 BACE molecules and 0.13 RMSE at 45 ESOL molecules, comparable to the entire foundation-model effect.

The split matters more than either. Slide 20 shows AUROC falling from 0.78 under a random split to 0.54 under a scaffold split. A 0.24 collapse from changing only the split, against the 0.01 to 0.03 we hope to measure.

The framing. Slide 41, titled "Molecular Pretraining: Graveyard of Good Ideas", carries the author's own annotation "OLD SLIDE!". The lecture's position is that the 2019-era pessimism is being revised, with gains small and objective-dependent. Our framing should match: given that gains are marginal and objective-dependent, is the corpus the lever? That is the one variable the whole benchmarking study never manipulates.

4. Does the corpus matter?

ChemFM (Cai et al., Commun. Chem. 2026)

The cleanest corpus comparison in print, and our reference [3]. A causal language model over SMILES at 1 billion and 3 billion parameters. Before the large run they used scaling experiments to choose the corpus: validation loss followed a clean power law in model size for UniChem but **saturated above about 60 million parameters for ZINC20**. They define an "informative" corpus operationally as one that fails to saturate.

They then confirmed it downstream. A 1-billion-parameter model pretrained on ZINC20, which has 1.8 billion molecules, loses to the same model pretrained on UniChem's 178 million molecules on 9 of 11 MoleculeNet tasks, several by large margins, with the remaining two essentially tied. A corpus ten times smaller wins.

Their explanation is redundancy: ZINC20 is a ligand-discovery library built from analogue series around mainstream medicinal-chemistry scaffolds, so its effective information content is far below its molecule count. They never measure that. No diversity statistic, no scaffold entropy, no redundancy metric is computed on either corpus. The claim is qualitative.

That omission is a gift. Our Stage A can supply exactly the measurement they assert, and our fixed-size composition experiment is the controlled version of their comparison, which confounds composition with a tenfold size difference.

Two caveats before we lean on it. Their scaling signal lives at 10 to 200 million parameters; our encoder is 8.7 million, below their smallest model, where no saturation is visible for either corpus. And the decisive evidence is a single run with no error bars where two of the eleven comparisons differ by 0.001, so quote it as "nine clear wins and two ties", not "9 of 11".

NaFM (Ding et al., Nat. Mach. Intell. 2026)

The natural-product foundation model, and the paper closest to our Stage D. A GIN pretrained on about 0.6 million COCONUT molecules with two objectives designed for natural products: reconstruction of masked scaffold subgraphs including topology, and contrastive learning weighted by scaffold similarity.

It beats every general-purpose pretrained baseline and is "the only pre-trained model that surpasses the rule-based ECFP fingerprint" on natural-product taxonomy classification.

The experiment we need is section 2.5. They re-pretrained three published models on the same COCONUT corpus and measured the effect:

Model	Change on re-pretraining with COCONUT
MolCLR	-4.3 AUPRC, low data
PretrainGNN	-6.5 AUPRC, high data
Mole-BERT	+3.8 high data, -5.3 low data

Two of three got worse. Their conclusion: "the observed advantages do not arise from dataset choice or data bias", the driver is the objective.

Read carefully, that experiment is confounded. Each model's original corpus was three to seventeen times larger than the COCONUT replacement, so the result is consistent with a pure scale effect rather than a domain effect. Our size-matched design is the clean version. Three consequences: a null or negative COCONUT result is the *expected* outcome and should be pre-registered as such; Stage D should be reported as a curve over fine-tuning set size, because Mole-BERT's effect changed sign with data density; and since NaFM evaluates on zero generic property tasks, the transfer question our Stage D asks is genuinely unanswered.

Their corpus preparation is undocumented beyond a preprocessed pickle file, so our documented COCONUT pipeline is itself a contribution.

"Throwing Everything AND the Kitchen Sink at CheMeleon" (West and Burns, OpenADMET, 4 August 2026)

The closest public precedent to Stage D, and the citable substitute for the Incyte poster that Yuxin described and that we could not find. The authors perturbed CheMeleon's pretraining extensively and found it remarkably robust: large dataset perturbations produced only marginal downstream gains, and pretraining loss turned out to be a poor proxy for fine-tuning performance. The only positive signals came from *relevant* molecules, with a 40,000-molecule set of analogues beating a diverse 10-million-molecule pool.

If that holds, our Stage D hypothesis sharpens: a natural-product corpus should help only on tasks whose molecules resemble natural products, and corpus size should matter far less than corpus-to-task proximity. That is testable with the Stage E machinery we already have.

Status. *Our automated reader was stopped by a model safety filter partway through this post, so the summary above rests on the reading list's description and the citation we verified. The full text is saved at* `papers/06_kitchen_sink_openadmet.txt` *and one of us should read it directly before the check-in. Note also that the reading list credits Pat Walters; the byline is Devany West and Jackson Burns.*

chemeleon-unlearn (Zalte)

Not yet retrieved; the GitHub repository was not found under the path in the reading list and needs a corrected link. The reading list reports the claim that re-pretraining CheMeleon on 200,000 molecules takes about one hour on a single RTX 4090 and matches the public 1-million-molecule model downstream. If true it de-risks our entire compute plan, and it is also an independent data point that the size curve flattens early. Worth chasing before Check-in 1.

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5. How much data is enough?

Chen et al., NeurIPS 2023 - neural scaling laws

Performance against *labelled* training-set size on HIV, MUV, PCBA and QM9, at nine nested fractions with five seeds, fitted as a power law in log-log space. No plateau appears anywhere in the explored range. Graph models and fingerprints are the most data-efficient modalities; SMILES transformers the least.

Two findings matter for Stage C. Pretrained models follow the same power law but with a **higher intercept and a lower exponent**: pretraining buys a head start, not a better slope. On PCBA the curves cross at about 40,000 labelled molecules, beyond which the pretrained model is *worse* than from-scratch. So our downstream tasks should stay well below that, and the honest measure of a corpus's value is a low-label curve rather than a single full-data score.

Crucially, they scale labelled data, not pretraining data, and name the pretraining-size axis as unexplored in their limitations. Stage C is the experiment they did not run.

For the fit itself: use seven sizes with per-size seed means, fit both a pure power law and a saturating form, compare residuals, and define "saturated" operationally, for instance as the last two sizes lying within the seed-to-seed spread of the one-million checkpoint. With two seeds the uncertainty on the exponent will be large, so put extra seeds at the three largest sizes where saturation is actually decided.

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6. Benchmarks and evaluation

MoleculeNet (Wu et al., Chem. Sci. 2018)

Source of BBBP and ESOL. Three things we did not know when we chose them.

ESOL's recommended split is random, not scaffold. Our scaffold-split ESOL departs from the benchmark protocol, which needs justifying or reporting alongside the random-split number. The paper's own appendix shows the same model scoring 0.31 RMSE under one configuration and 0.97 in the benchmark table, so ESOL is highly sensitive to budget and split.

MoleculeNet's scaffold splits are deterministic, so their reported standard deviations exclude split variance entirely. Our seed-varying scaffold split is stricter than the published protocol, and we should report within-split and across-split variability separately.

The empty Bemis-Murcko scaffold is never discussed. Handling the 317 acyclic ESOL molecules that share it is therefore our contribution, not a known fix. Three options: a secondary key so acyclic molecules distribute across folds, proportional allocation of that block, or reporting metrics with and without it.

On BBBP, every model scores about 0.95 validation AUROC against 0.70 test, so differences between corpora will be compressed into a narrow band. Lipophilicity (4,200 molecules, regression) and BACE (1,513, scaffold, classification) are the cleaner instruments in the same size class; ClinTox, SIDER, MUV and FreeSolv should be avoided as primary tasks.

Therapeutics Data Commons (Huang et al., Nat. Chem. Biol. 2022)

Where our additional tasks should come from. The ADMET group protocol is the one to copy exactly: a 7:1:2 scaffold split with the **test set fixed** and only train and validation reshuffled across five runs, reporting mean and standard deviation. That single detail may explain most of our ESOL pain: our test mean moved 0.94 log units because we re-split the test set per seed.

Metric selection is per endpoint: AUROC for balanced binary data, AUPRC when positives are rare, mean absolute error for regression and Spearman correlation where the trend matters more than absolute error.

Concrete candidates, all scaffold-split:

Task	Size	Type	Metric	Why
Caco2_Wang	906	regression	MAE	permeability, drug-like, minutes to fine-tune
HIA_Hou	578	classification	AUROC	absorption, chemistry relevant to natural products
CYP3A4_Veith or CYP2D6_Veith	12-13k	classification	AUPRC	drawn from PubChem BioAssay, so maximal overlap with our PubChem size curve
LD50_Zhu	7,385	regression	MAE	environmental and industrial chemistry, far from ZINC, a distance probe

The CYP suggestion is the sharpest: it is the task family where pretrained models beat descriptor baselines by the widest margin in TDC's own table (0.739 against 0.616 AUPRC), and it is PubChem-derived, so our size curve should amplify any corpus effect there if one exists.

TDC's own headline is a warning: "the ML SOTA models do not work well consistently for these novel realistic endpoints. In some cases, methods based on learned features are worse than the efficient domain features."

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7. The natural-product-likeness filter

Ertl, Roggo and Schuffenhauer, JCIM 2008

The score behind our `zinc_nplike` arm. A naive-Bayes model over atom-centred fragments and fragment pairs, trained on 115,590 deglycosylated aglycons from the 2006 Dictionary of Natural Products against 290,000 commercially available synthetic compounds. Cross-validated ROC-AUC 0.977.

Four facts change how we must use and report it.

No numeric threshold is published. The paper refers only to "the optimal cutoff suggested by the ROC curve". RDKit's effective zero boundary is a convention, not Ertl's number. We must choose a cutoff explicitly and show sensitivity at 0, +0.5 and +1.0.

A model using only seven simple properties performs almost as well. LogP, polar surface area, heavy-atom count, nitrogen plus oxygen count, aromatic atoms, stereocentres and rotatable bonds nearly reproduce the fragment model. So `zinc_nplike` is confounded with all of those, and any downstream effect may be a property-distribution effect rather than a natural-product effect. A property-matched ZINC control

corpus is the obvious remedy.

Sugars were stripped before training, so the score measures aglycon scaffolds. Glycoside-rich COCONUT entries are part of our 23 percent sub-zero fraction.

The reference set is from 2006, fifteen years before COCONUT, so reference drift explains part of that fraction too.

And the authors' own warning, which belongs in our limitations section: the score "cannot be used as a criterion for the quality of a library, neither it is possible to conclude from it anything about the probability of bioactivity". Caffeine scoring -1.09 while losartan scores -0.97 is consistent with the construction, since the negative class was a catalogue of purchasable compounds and caffeine is in it. It is not a bug, and we should say so rather than apologising for it.

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8. Where the literature agrees

Six claims are supported by more than one independent source.

1. **Pretraining gains on small molecular tasks are small and inconsistent.** Hu 2020, Sun 2022, Deng 2023, Praski 2025 and the October lecture all agree.
2. **The objective matters more than the corpus, on current evidence.** Praski's four-above-ECFP pattern, NaFM's section 2.5 and the lecture's model ranking all point the same way. Nobody has tested it with the corpus as the only variable.
3. **Gains live in the low-data regime and vanish as labels accumulate.** Chen's crossover at 40,000 labels, the lecture's convergence by 847 BACE molecules, Deng's "competitive only below 1,000", Hu's largest gains on the smallest datasets.
4. **Evaluation variance swamps the effects being measured.** Deng's 30 splits, Sun's learning-rate reversal, Hu's ±4 to ±5 seed deviations, the lecture's critical difference of 1.61 to 2.38 rank points.
5. **Fixed descriptors and fingerprints remain hard to beat.** Praski's ECFP result, Deng's random forest, TDC's own baselines, the lecture's descriptor-versus-no-descriptor gap.
6. **Nobody has run a size-matched corpus comparison at a fixed recipe.** ChemFM confounds composition with a tenfold size difference; NaFM confounds it with a three-to-seventeen-fold reduction; Sun varied scale only; Hu, CheMeleon, Chen and Praski did not vary the corpus at all.

Point six is our project.

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9. What this means stage by stage

Stage A should now produce the measurements ChemFM asserted but never made: internal diversity, sphere-exclusion diversity at two sample sizes so its decay exposes analogue-series redundancy, scaffold entropy and concentration, and the seven Ertl properties with standardised effect sizes between corpora. That makes "the corpora differ" quantitative rather than visual, and it is publishable regardless of what the GPU stages produce.

Stage B gains a cheap and strong instrument from CheMeleon's own paper: k-nearest-neighbour probing of the frozen representation, which needs almost no GPU and directly tests whether a corpus organises nearby test molecules better.

Stage C should fit both a power law and a saturating form, define saturation operationally against the seed spread, and keep the one-million checkpoint as an off-curve reference point rather than a fitted point unless the implementation question is resolved in favour of the original code. The flat-curve outcome is likely, given Sun's 500-fold null and the claim from chemeleon-unlearn, and it is a result worth reporting cleanly.

Stage D should expect a null. Pre-register the effect size of interest, report a low-label curve rather than a single number, and keep the property-matched ZINC control in reserve to separate "natural-product

chemistry" from "larger, more polar, more stereocentres".

Stage E is where the Kitchen Sink finding would show up if it is real: bin test molecules by similarity to the corpus and look for the gain to concentrate in the near bins. Expect non-monotonicity, because CheMeleon's own authors report that the representation collapses similar molecules together.

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10. Changes to make before the first pretraining run

Ordered by how much they cost us if skipped.

1. **Fix the test split and reshuffle only train and validation,** following TDC. This is probably the single largest source of our ESOL noise.
2. **Raise the seed count.** Two seeds cannot separate effects of the size the literature reports. Deng needed 30 splits; Hu used 10. Put the extra seeds at the sizes where saturation is decided rather than spreading them evenly.
3. **Freeze the Mordred column set and scaler from the released one-million store,** which simultaneously settles our open issue and matches what the published checkpoint was standardised against.
4. **Add a strong fingerprint baseline:** ECFP *count* vectors with a tree model, which is the one baseline Praski's authors call mandatory, plus the graph-free feed-forward network on Mordred descriptors that Chemprop supports natively.
5. **Demote ESOL to a sanity check** and add two or three TDC tasks, with Caco2 and HIA as the cheap pair and a CYP endpoint as the PubChem-overlap probe.
6. **Report per-task, not pooled.** Separate classification from regression and flag negative transfer per dataset, as Hu's shaded cells do.
7. **Deduplicate corpora against every downstream test set by InChIKey** and report near-duplicate counts at Tanimoto 0.9. Our current exact-SMILES count is a lower bound and we already know it.
8. **Record corpus metadata with every run:** source, size, diversity statistics, overlap counts. No paper in this review does this, and it is what makes a corpus-versus-performance plot possible at the end.

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11. What we can claim as new

- A size-matched, fixed-recipe comparison of corpus composition. Nobody has one.
- A measured diversity and redundancy comparison between pretraining corpora,

which ChemFM asserts qualitatively and Praski omits entirely.

- The pretraining-corpus size axis, named as unexplored in Chen's limitations.
- Whether natural-product pretraining transfers to non-natural-product

endpoints, which NaFM never tests.

- Whether pretraining loss predicts downstream gain across corpora, which

CheMeleon explicitly leaves to future work.

- Benchmark-exposure accounting per corpus arm, absent from CheMeleon, ChemFM

and NaFM alike.

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12. References

#	Work	Venue	Role here
1	Burns, Zalte et al., CheMeleon	JCIM 2026	our fixed recipe
2	Heid et al., Chemprop	JCIM 2024	architecture, baselines
3	Sun, Dai, Yu	NeurIPS 2022	methodological model, corpus scale null
4	Cai et al., ChemFM	Commun. Chem. 2026	corpus-only comparison

5	Ding et al., NaFM	Nat. Mach. Intell. 2026	natural-product pretraining
6	West and Burns, Kitchen Sink	OpenADMET blog 2026	closest Stage D precedent
7	Ertl, Roggo, Schuffenhauer	JCIM 2008	NP-likeness filter
8	Huang et al., TDC	Nat. Chem. Biol. 2022	tasks and protocol
9	Wu et al., MoleculeNet	Chem. Sci. 2018	tasks and their limits
10	Chen et al.	NeurIPS 2023	scaling-law method
11	Deng et al.	Nat. Commun. 2023	evaluation audit
12	Hu et al.	ICLR 2020	negative transfer
13	Praski, Adamczyk, Czech	arXiv 2508.06199	28-model frozen benchmark
14	Barzilay and Xie	6.3930 lecture, 1 Oct 2026	the study we extend

PDFs are in `papers/`, named by the numbers above where they were retrievable. Two could not be downloaded as PDFs and are noted in place: the Chemprop article (open-access text saved instead) and the chemeleon-unlearn report.